SYNTHETIC DEMO DATA ? NOT A REAL INSPECTION RECORD

WIND TURBINE PERIODIC INSPECTION REPORT

Asset ID: WT-DEMO-014
Manufacturer: Northstar Renewable Systems (fictional)
Model: NSR-4.2-Demo
Inspection date: 2026-10-18
Location: Wind Farm Alpha, North Array, Turbine 14 (fictional)

Inspection scope

Visual inspection of external blade surfaces, nacelle equipment, gearbox observation points, accessible fasteners, lubrication indicators, and supervisory vibration trends.

Inspection findings

The tower and accessible bolted connections showed no visible deformation. Protective coating was intact except for two small areas of superficial abrasion near the lower access platform. No active corrosion was observed.

Blade condition

Blade A showed a 35 mm superficial leading-edge coating chip approximately 8.2 m from the root reference. Blades B and C showed normal surface wear for the fictional service interval. No laminate exposure or lightning receptor damage was observed.

Gearbox condition

Oil level was within the marked operating range. The sight glass was clear. No leakage was visible at accessible seals. The fictional laboratory oil sample is pending and is not included in this demo report.

Vibration observation

The reviewed synthetic trend remained below the fictional alert threshold. A small increase at the high-speed shaft frequency was observed but was not sufficient to classify as a defect. Continue trend monitoring.

Recommendations

1. Repair the Blade A coating chip during the next suitable low-wind maintenance window.
2. Reinspect the two tower coating abrasions within 90 days.
3. Review the pending fictional oil analysis before returning to the normal annual cycle.
4. Continue monthly vibration trending.

Inspector: Morgan Lee (fictional)
Next inspection date: 2027-04-18

This document is synthetic, contains no proprietary information, and is provided only to demonstrate AssureAI Milestone 1 document ingestion.